

Warm Up:

1. Simplify: $5(2x + 3) - 4x$
→ $10x + 15 - 4x = 6x + 15$
2. Evaluate: $|3 - 5x|$, if $x = 2$
→ $|3 - 10| = |-7| = 7$
3. Round 4.267 to the nearest hundredth.
→ 4.27
4. A car travels 260 miles on 8 gallons. Find the miles per gallon.
→ $260 \div 8 = 32.5$ miles per gallon
5. A team scored 1848 points over 22 games. Find the points per game.
→ $1848 \div 22 = 84$ points per game

1.5 Homework Questions?

Speak Like A Mathematician:

descriptive modeling: Using a mathematical model – a graph, table, formula, or diagram – to describe a real situation.

metric: A rule or measurement used to compare or rank things numerically.

accuracy: How close a measured or calculated value is to the true value.

Example 1: Descriptive Modeling

- A school store tracks hoodie sales for five months: 42, 55, 38, 61, and 49 hoodies. Describe two different mathematical models the store could use to summarize these sales, and state what each one tells you.
 - A bar graph shows month-to-month comparison; the mean (49 hoodies) summarizes a typical month. A table preserves every value; a graph shows the trend. Each model keeps some information and hides some.
- Which model would be most useful for deciding how many hoodies to order next month? Explain.
 - The graph or the trend – a single mean hides that sales swing between 38 and 61, so ordering exactly 49 would run short in a strong month.

Example 2: Compare Metrics

- Two players' season statistics are shown. Use two different metrics to decide who had the better season.

Player	Points	Games
Alvarez	1,848	22
Brooks	1,530	17

→ Total points: Alvarez leads, 1848 to 1530. Points per game: Alvarez $1848 \div 22 = 84$; Brooks $1530 \div 17 = 90$ – Brooks leads. The metric you choose determines the answer.

This is the whole point of the lesson: two honest metrics can rank the same data differently. Always say WHICH metric you used.

Example 3: Choose a Metric

- A principal wants to identify the “most improved” class on a state exam. Class A rose from 62 to 74. Class B rose from 80 to 88. Which class improved more? Defend your choice with a metric.
 - Raw gain: A improved 12 points, B improved 8 – A wins.
 - Percent gain: A is $12/62 \approx 19.4\%$, B is $8/80 = 10\%$ – A still wins.
 - Both metrics agree here, which makes the conclusion strong.

Example 4: Accuracy

- A digital thermometer reads 98.5°F when the true temperature is 98.6°F . Find the accuracy of the reading.
 - $|98.5 - 98.6| = 0.1^{\circ}\text{F}$ – the reading is accurate to within one tenth of a degree.
- A bag of flour is labeled 5 pounds. A scale shows 4.92 pounds. Find the accuracy, and express it as a percent of the labeled weight.
 - $|4.92 - 5| = 0.08$ pounds; $0.08 \div 5 = 0.016 = 1.6\%$ off the labeled weight.

Accuracy uses absolute value – it is a distance from the true value, so it is never negative. That is 1.5 doing work here in 1.6.

Tonight's Homework: Paper Practice 1.6 (#1-10)